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GATE ECE · ELECTRONIC DEVICES AND VLSI · CHAPTER-WISE PYQS

BJT and FET

GATE ECE Electronic Devices and VLSI · Previous Year Questions with Answer Key

29	18	1992 – 2026	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

INSIDE THIS PDF

Marks 1	19 questions
Marks 2	10 questions
Answer key	at the end

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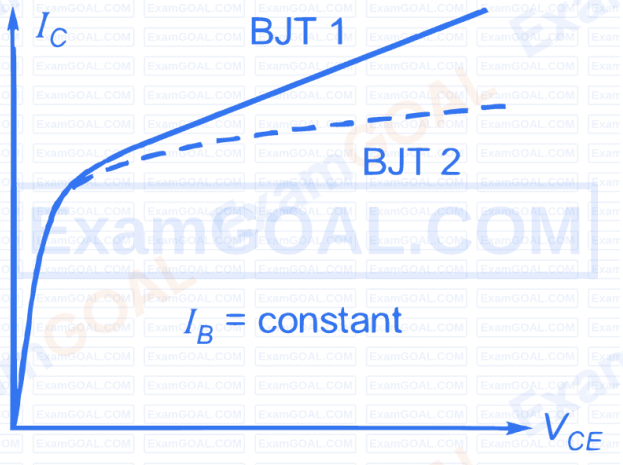
Marks 1 19 questions

Q1. GATE ECE 2026

Open solution

Figure shows the output characteristics of two different Bipolar Junction Transistors (BJT), BJT 1 with magnitude of Early voltage $|V_{A1}|$, and BJT 2 with magnitude of Early voltage $|V_{A2}|$.

Which of the following options is/are correct regarding the Early voltages?



(A) $|V_{A1}| > |V_{A2}|$

(B) $|V_{A1}|$ is infinitely large

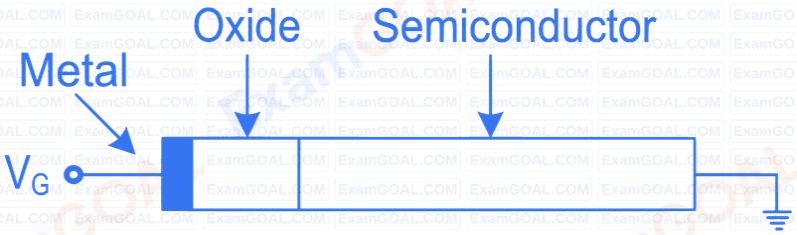
(C) $|V_{A1}| < |V_{A2}|$

(D) $|V_{A2}|$ is finite

Q2. GATE ECE 2022

Open solution

An ideal MOS capacitor (p-type semiconductor) is shown in the figure. The MOS capacitor is under strong inversion with V_G = 2V. The corresponding inversion charge density (Q_{IN}) is 2.2 μC/cm^2. Assume oxide capacitance per unit area as C_{ox} = 1.7 μF/cm^2. For V_G = 4V, the value of Q_{IN} is _____ μC/cm^2 (rounded off to one decimal place).



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Q3. GATE ECE 2017 Set 2

An npn bipolar junction transistor (BJT) is operating in the active region. If the reverse bias across the base-collector junction is increased, then

(A) the effective base width increases and common – emitter current gain increases

(B) the effective base width increases and common – emitter current gain decreases

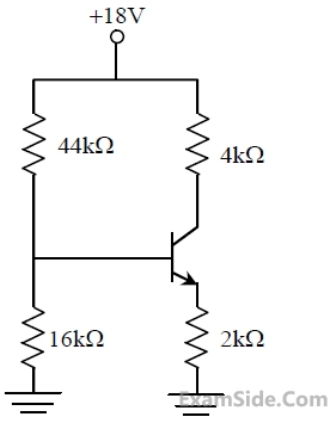
(C) the effective base width decreases and common – emitter current gain increases

(D) the effective base width decreases and common – emitter current gain decreases

Open solution

Q4. GATE ECE 2017 Set 2

Consider the circuit shown in the figure. Assume base-to-emitter voltage V_{BE}=0.8 V and common base current gain (α) of the transistor is unity.

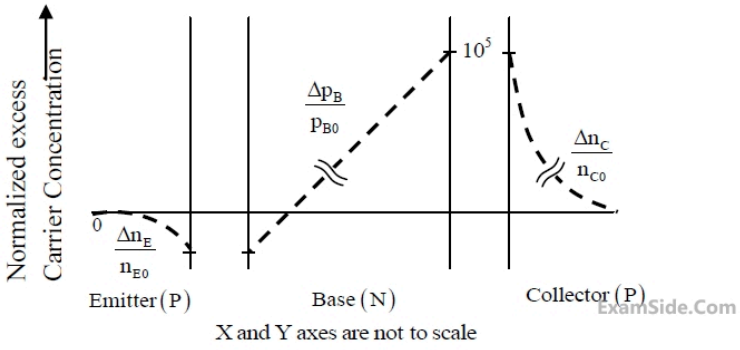


The value of the collector-to–emitter voltage V_{CE} (in volt) is _____.

Open solution

Q5. GATE ECE 2017 Set 1

For a narrow base PNP BJT, the excess minority carrier concentration ($\triangle n_E$ for emitter, $\triangle p_B$ for base, $\triangle n_E$ for collector) normalized to equilibrium minority carrier concentration ($\triangle n_{E0}$ for emitter, $\triangle p_{B0}$ for base, $\triangle n_{C0}$ for collector) in the quasi-neutral emitter, base and collector regions are shown below. Which one of the following biasing modes is the transistor operating in?



(A) Forward active (B) Saturation (C) Inverse active (D) Cutoff

Open solution

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Q6. GATE ECE 2016 Set 2

The Ebers-Moll model of a BJT is valid

(A) only in active mode

(B) only in active and saturation modes

(C) only in active and cut-off modes

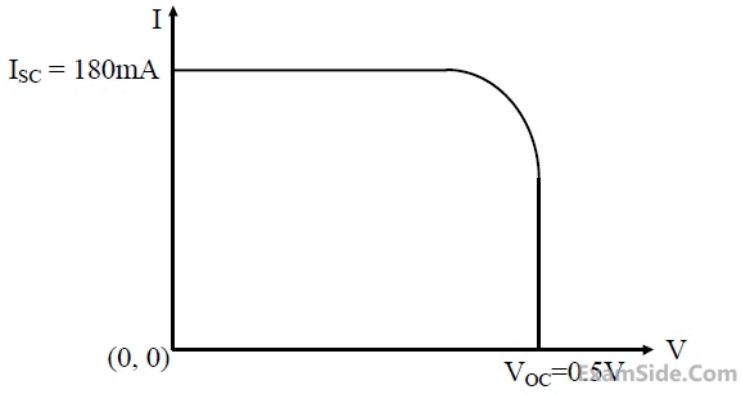
(D) in active, saturation and cut-off modes

Open solution

Q7. GATE ECE 2016 Set 3

The figure shows the I-V characteristics of a solar cell illuminated uniformly with solar light of power 100 mW/cm^2. The solar cell has an area of 3 cm^2 and a fill factor of 0.7. The maximum efficiency (in %) of the device is _____.

Open solution



Q8. GATE ECE 2015 Set 3

In the circuit shown in the figure, the BJT has a current gain (β) of 50. For an emitter-base voltage V_{EB} = 600 mV, the emitter-collector voltage V_{EC} (in Volts) is _____.

Open solution

Q9. GATE ECE 2015 Set 3

If the base width in a bipolar junction transistor is doubled, which one of the following statements will be TRUE?

(A) Current gain will increase

(B) Unity gain frequency will increase

(C) Emitter base junction capacitance will increase

(D) Early voltage will increase

Open solution

Q10. GATE ECE 2014 Set 2

An increase in the base recombination of a BJT will increase

(A) the common emitter dc current gain β

(B) the breakdown voltage BV_{CEO}

(C) the unity-gain cut-off frequency f_T

(D) the transconductance g_m

Open solution

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Q11. GATE ECE 2011

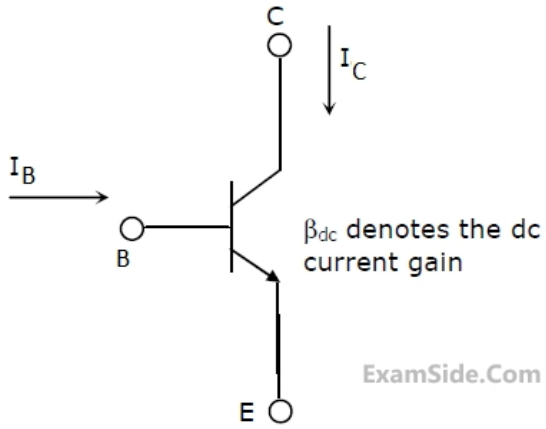
For a BJT the common base current gain α = 0.98 and the collector base junction reverse bias saturation current I_{CO} = 0.6μA. This BJT is connected in the common emitter mode and operated in the active region with a base drive current I_B=20μA. The collector current IC for this mode of operation is

(A) 0.98mA (B) 0.99mA (C) 1.0mA (D) 1.01mA

Open solution

Q12. GATE ECE 2002

If the transistor in Figure is in saturation, then



(A) I_C is always equal to $b_{dc}I_B$ (B) I_C is always equal to $-b_{dc}I_B$

(C) I_C is greater than or equal to $b_{dc}I_B$ (D) I_C is less than $b_{dc}I_B$

Open solution

Q13. GATE ECE 2001

MOSFET can be used as a

(A) Current controlled capacitor (B) Voltage controlled capacitor

(C) Current controlled inductor (D) Voltage controlled inductor

Open solution

Q14. GATE ECE 1995

The breakdown voltage of a transistor with its base open is BV_{CEO} with emitter open is BV_{CBO}, then

(A) $BV_{CEO} = BV_{CBO}$ (B) $BV_{CEO} > BV_{CBO}$

(C) $BV_{CEO} < BV_{CBO}$ (D) BV_{CEO} is not related to BV_{CBO}

Open solution

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Q15. GATE ECE 1995

A BJT is said to be operating in the saturation region if

(A) both the junctions are reverse biased

(B) base-emitter junction is reverse biased and base-collector junction is forward biased.

(C) base-emitter junction is forward biased and base-collector junction reverse biased

(D) both the junctions are forward biased

Open solution

Q16. GATE ECE 1995

The Ebers-Moll model is applicable to

(A) bipolar junction transistors

(B) NMOS transistors

(C) unipolar junction transistors

(D) junction field-effect transistors

Open solution

Q17. GATE ECE 1995

The Early-Effect in a bipolar junction transistor is caused by

(A) Fast-turn-on.

(B) Fast-turn-off.

(C) Large collector-base reverse bias

(D) Large emitter-base forward bias

Open solution

Q18. GATE ECE 1994

The transit time of the current carriers through the channel of an FET decides its _____characteristics.

Open solution

Q19. GATE ECE 1994

Channel current is reduced on application of a more positive voltage to the gate of a depletion mode n-channel MOSFET.

(A) TRUE

(B) FALSE

Open solution

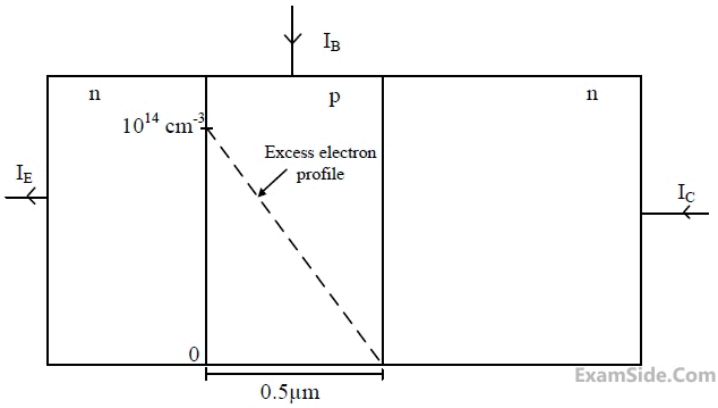
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Marks 2 10 questions

Q20. GATE ECE 2016 Set 3

Open solution

The injected excess electron concentration profile in the base region of an npn BJT, biased in the active region, is linear, as shown in the figure. If the area of the emitter-base junction is 0.001 cm^2, μ_n = 800 cm^2/(V-s) in the base region and depletion layer widths are negligible, then the collector current I_C (in mA) at room temperature is _____. (Given: thermal voltage V_T = 26 mV at room temperature, electronic charge q = $1.6 \times 10^{-19}\ C$)



Q21. GATE ECE 2015 Set 3

Open solution

An npn BJT having reverse saturation current $I_s = 10^{-15}\ A$ is biased in the forward active region with V_{BE} = 700 mV. The thermal voltage (V_T) is 25 mV and the current gain (β) may vary from 50 to 150 due to manufacturing variations. The maximum emitter current (in μA) is _____.

Q22. GATE ECE 2014 Set 4

Open solution

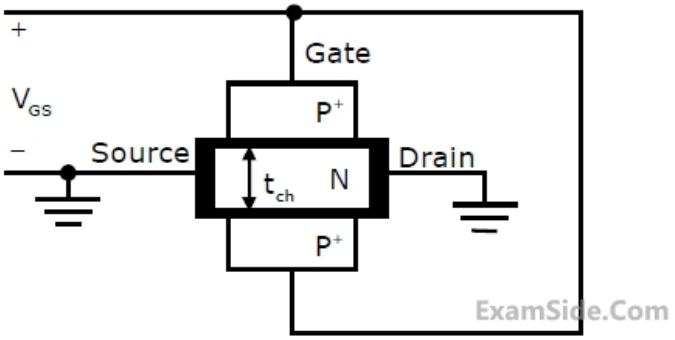
Consider two BJT's biased at the same collector current with area A_1 = 0.2 μm × 0.2 μm and A_2 = 300 μm × 300 μm. Assuming that all other device parameters are identical, kT/q = 26 mV, the intrinsic carrier concentration is 1 × 10^{10} cm^{-3}, and q = 1.6 × 10^{-19} C, the difference between the base-emitter voltages (in mV) of the two BJT's (i.e., $V_{BE1} - V_{BE2}$) is_____.

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Q23. GATE ECE 2011

Open solution

The channel resistance of an N-channel JFET shown in the figure below is 600 W when the full channel thickness (t_{ch}) of 10 μm is available for conduction. The built-in voltage of the gate P^+N junction (V_{bi}) is -1 V. When the gate to source voltage (V_{GS}) is 0 V, the channel is depleted by 1 μm on each side due to the built in voltage and hence the thickness available for conduction is only 8 μm



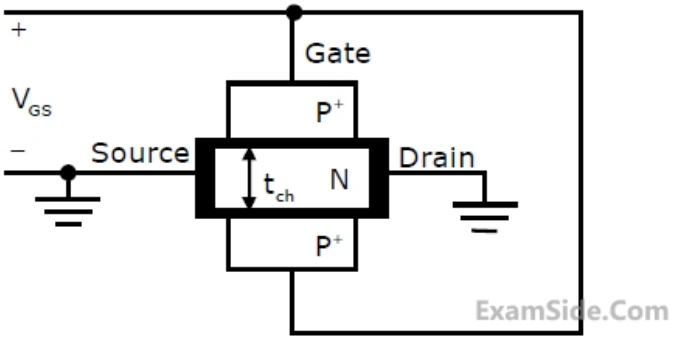
The channel resistance when V_{GS} = -3 V is

(A) 360 Ω (B) 917 Ω (C) 1000 Ω (D) 3000 Ω

Q24. GATE ECE 2011

Open solution

The channel resistance of an N-channel JFET shown in the figure below is 600 W when the full channel thickness (t_{ch}) of 10 μm is available for conduction. The built-in voltage of the gate P^+N junction (V_{bi}) is -1 V. When the gate to source voltage (V_{GS}) is 0 V, the channel is depleted by 1 μm on each side due to the built in voltage and hence the thickness available for conduction is only 8 μm



The channel resistance when V_{GS} = 0 V is

(A) 480 Ω (B) 600 Ω (C) 750 Ω (D) 1000 Ω

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Q25. GATE ECE 2010

In a uniformly doped BJT, assume that N_E, N_B and N_C are the emitter, base and collector dopings in atoms/cm^3, respectively. If the emitter injection efficiency of the BJT is close to unity, which one of the following conditions is TRUE?

Open solution

(A) $N_E = N_B = N_C$

(B) $N_E \gg N_B$ and $N_B > N_C$

(C) $N_E = N_B$ and $N_B < N_C$

(D) $N_E < N_B < N_C$

Q26. GATE ECE 2007

The DC current gain (β) of a BJT is 50. Assuming that the emitter injection efficiency is 0.995, the base transport factor is:

Open solution

(A) 0.980 (B) 0.985 (C) 0.990 (D) 0.995

Q27. GATE ECE 1999

An n-channel JEFT has I_{DSS} = 2 mA and V_p = −4 V. It's transconductance g_m (in mA/V) for an applied gate-to-source voltage V_{GS} of −2V is:

Open solution

(A) 0.25 (B) 0.5 (C) 0.75 (D) 1.0

Q28. GATE ECE 1992

In a transistor having finite B, the forward bias across the base emitter junction is kept constant and the reverse bias across the collector base junction is increased. Neglecting the leakage across the collector base junction and the depletion region generating current, the base current will _____. (increase/decrease/remain constant).

Open solution

Q29. GATE ECE 1992

An n-channel JFET has a pinch-off voltage V_P = -5 V, V_{DS}(max) = 20 V and g_m = 2 mA/V. The minimum 'ON' resistance is achieved in the JEFT for

Open solution

(A) V_{GS} = - 7 V and V_{DS} = 0 V

(B) V_{GS} = 0 V and V_{DS} = 0 V

(C) V_{GS} = 0 V and V_{DS} = 20 V

(D) V_{GS} = - 7 V and V_{DS} = 20 V

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Answer Key

Marks 1

1. **C, D**	2. **5.6**	3. **C**	4. **6**	5. **C**
6. **D**	7. **21**	8. **2**	9. **D**	10. **B**
11. **D**	12. **D**	13. **B**	14. **C**	15. **D**
16. **A**	17. **C**	18. **Drain**	19. **B**	

Marks 2

20. **6.656**	21. **1475**	22. **381**	23. **C**	24. **C**
25. **B**	26. **B**	27. **B**	28. **Decrease**	29. **C**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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